

MicroPulse

Training Monotony & Strain — Coach Guide

The original load-monitoring math (Foster 1998) — variability matters as much as volume

1. Overview — what is Monotony & Strain?

Training Monotony & Strain is a per-player overtraining-risk monitor that has been in continuous use since 1998 — long before GPS existed. It answers a question that volume alone cannot: how variable is the player's training load from one day to the next? A player who trains at 600 AU every single day is at higher overtraining risk than one who alternates 400 / 800, even though both have the same weekly total.

Foster's original paper (*Med Sci Sports Exerc*, 1998) established the relationship between consistent high-load weeks and the onset of illness, soft-tissue injury and overtraining syndrome. The framework has since been validated by AC Milan, Real Madrid, and dozens of other elite organisations.

Why this card matters for Lite teams

Foster Monotony & Strain needs only sRPE + duration — two numbers a player can log in 10 seconds after every session. It does NOT require Catapult B2-3 efforts, IMA Free Running, or any premium parameter. This is the single highest-quality training-load signal available to a club without a Pro Catapult plan.

Where do you find this card?

- Page: /coach/load-intelligence — appears as a dedicated card on the Load Intelligence page.
- Visible to all Catapult plans (Lite and Pro / Premium). It's a strong cross-check even if you have higher-tier data.

2. The math — exactly how it's computed

Foster's framework is intentionally simple — five numbers, all derived from the past 7 days of training:

Term	Formula	Plain-English meaning
daily_load	$\text{sRPE} \times \text{duration (minutes)}$	How hard one session was, multiplied by how long it lasted. A 30-min RPE 8 session = 240 AU.
weekly_load	Sum of daily_load over the past 7 days	Total training volume for the week, in AU.
mean_load	$\text{weekly_load} \div 7$	Average daily load across the week (including rest days as 0).
sd_load	Standard deviation of daily_load across 7 days	How spread out the daily loads are. High SD = some hard days, some easy days. Low SD = every day is the same.
MONOTONY	$\text{mean_load} \div \text{sd_load}$	The headline ratio. High monotony = same load every day = no recovery rhythm.

Term	Formula	Plain-English meaning
STRAIN	$\text{weekly_load} \times \text{MONOTONY}$	Combines volume and monotony into one number. High volume + high monotony = elite-level danger zone.

Why both numbers matter

Volume on its own is misleading. Two players with identical 4000 AU weeks can carry very different injury risk: one trained 4 days at 1000 AU each (monotony ≈ 1.0 , strain ≈ 4000), the other trained 7 days at 571 AU each (monotony $\approx \infty$, strain in the danger zone). Foster's insight was that **the body recovers from spikes more easily than it recovers from chronic same-ness**.

3. Two variants — RPE vs Volume fallback

The card automatically picks the best input it has access to. The badge at the top of the card tells you which variant is currently active.

Variant A — RPE $\times$ duration (canonical, green badge)

This is Foster's original method. Triggered when at least 5 distinct days have RPE entries logged across the squad in the 7-day window. The numbers you see are directly comparable to published Foster thresholds (≤ 1.5 SAFE, > 2.0 HIGH; ≤ 4500 SAFE, > 6000 DANGER).

Variant B — Volume (m) fallback (blue badge)

When RPE compliance is too low (< 5 distinct days in the window), the card switches to a volume-only variant using GPS total distance. Total distance is divided by 10 so a typical 6000 m session becomes "600 AU" — close enough to the RPE-load range that the same thresholds remain approximately valid.

Variant trade-off

Volume-only is a weaker signal because it ignores subjective effort — a player can complete a high-distance session feeling fresh, or a low-distance session feeling crushed. RPE captures that. If your team is consistently in the volume variant, push for better RPE compliance — even 1 player logging every day would meet the threshold.

4. Column-by-column glossary

The card shows a table with one player per row, sorted by Strain descending (highest danger first).

Column	What it shows	How to read it
Player	Player name and position.	Self-explanatory.
Days	Number of days with training data in the 7-day window (X/7).	If a player has 0/7 they don't appear in the table. Players with 1-2/7 show "—" for monotony / strain (need at least 2 days with non-zero data).
Weekly load	Sum of daily loads in the 7-day window. AU when in RPE variant; scaled distance when in Volume variant.	The pure volume number. Compare across players: who has done the most this week.

Column	What it shows	How to read it
Monotony	$\text{mean_load} \div \text{sd_load}$. Measures day-to-day similarity.	≤ 1.5 SAFE (green), 1.5-2.0 WATCH (yellow), > 2.0 HIGH (red). Colour-coded directly.
Strain	$\text{weekly_load} \times \text{monotony}$. Combined volume $\times$ variability.	≤ 4500 SAFE (green), 4500-6000 WATCH (yellow), > 6000 DANGER (red). Colour-coded directly.
Status	Combined badge — worst of the two flags.	What you read first to triage who needs attention.

5. Status flags — what each one means

Status	Triggered when	What it means	What you should do
DANGER	Strain > 6000 OR Monotony > 2.0	Foster's "illness/injury danger zone". Either the volume $\times$ variability product is dangerously high, or there is no recovery rhythm at all.	Insert a hard recovery day or full rest within 48 hours. If multiple players are in DANGER simultaneously, the team has a periodisation problem — review the week structure.
WATCH	Strain 4500-6000 OR Monotony 1.5-2.0	Trending toward danger zone. Sustainable for a week or two but should not stay here.	Build at least one clearly easy day into the next 7 days. Re-check after 3 days — has the player moved into SAFE or progressed to DANGER?
SAFE	Strain ≤ 4500 AND Monotony ≤ 1.5	Healthy week structure. Volume is sustainable and there is genuine variation between days.	No specific action. Keep the week structure.
—	Fewer than 2 days with non-zero data in the window	Not enough data to compute monotony reliably (would divide by zero).	Wait until the player has trained 2+ days in the last week before trusting the card.

6. Scenarios — reading the combinations

Scenario A: High volume + low monotony (Strain SAFE, Monotony SAFE)

Looks like: Weekly load 4500 AU, monotony 1.2, strain 5400. **What it means:** Player worked hard but had real variability — probably 2 hard days, 2 medium days, 3 easy/rest days. This is exactly the structure the body recovers from. No action needed.

Scenario B: Moderate volume + high monotony (Strain WATCH, Monotony HIGH)

Looks like: Weekly load 3500 AU, monotony 2.4, strain 8400. **What it means:** Player trained 6-7 days at the same medium load. The volume isn't extreme but the lack of variation is. This is the classic overtraining setup — a player who never has a clearly easy day. **Action:** Schedule a deliberately light session (< 200 AU) within 48 hours. Monotony will drop fast.

Scenario C: High volume + high monotony (Strain DANGER, Monotony HIGH)

Looks like: Weekly load 5500 AU, monotony 2.6, strain 14300. **What it means:** Worst case. High volume AND no variability. Foster's data suggests this is where illness, soft-tissue injury and overtraining syndrome cluster. **Action:** Mandatory full rest day within 24 hours, then a planned easy session before the next hard work. Talk to the player about wellness markers — sleep, soreness, mood.

Scenario D: Low volume + high monotony (Strain SAFE, Monotony WATCH/HIGH)

Looks like: Weekly load 1200 AU, monotony 2.1, strain 2520. **What it means:** Player is doing very little training but what little they do is at the same intensity every day. Common in returning-from-injury situations or when a player is being eased back. **Action:** Monotony alone isn't the problem at this volume — Foster's risk thresholds were derived for full training loads. Use this as a signal to start adding variation back in as volume increases.

7. Workflow — when to use this card

Daily — morning load review (3 min)

- Open the card on your way to morning planning. Sort is already by Strain descending — top rows need attention first.
- Players in DANGER: do not put them through hard work today. Either rest, technical / scheme work, or low-load recovery session.
- Players in WATCH: avoid stacking another hard day. If yesterday was hard, today should be easy or off.

Weekly — after match-week planning (10 min)

- Run through the squad before publishing the next week's plan. Are any positional groups (e.g. all your defenders) clustered in WATCH? That suggests a programming pattern.
- Cross-reference with HSR Intelligence and MD Comparison — same player flagged across multiple cards is a stronger signal than any one card alone.
- If a player is consistently DANGER for 2+ weeks running, escalate to the medical team for a comprehensive screen (sleep, RHR, jump tests, blood work if available).

Pre-season — RPE compliance check

- First two weeks of pre-season are the time to fix RPE habits. The card will sit in Volume variant until RPE is logged 5+ days/week.
- Make RPE entry part of the post-session ritual — 10 seconds before players leave the dressing room.

8. Frequently asked questions

Why does a player I know is overtrained show SAFE?

Foster Monotony & Strain measures the past 7 days only — it does not see chronic accumulation across multiple weeks. A player who has been DANGER for 4 weeks straight, then took a recovery week, will show SAFE this week even though they are still in a deep hole. Cross-check Quadrant View (chronic load axis) and player-specific wellness scores to catch this.

Why is monotony "—" for a player who clearly trained?

The card needs at least 2 days with non-zero load AND a standard deviation above 0.5 to compute monotony — otherwise the formula divides by zero or produces a misleading infinity. A player who trained the exact same amount on every day they trained will also show "—" for this reason. This is a feature, not a bug — it stops the card from showing nonsense numbers.

Can I trust the Volume variant the same way as the RPE variant?

Roughly, yes — the thresholds are calibrated to be approximately equivalent (typical 6000 m session $\approx$ 600 AU). But subjective effort is genuinely informative and the Volume variant misses it. A 6000 m session at RPE 5 is very different from a 6000 m session at RPE 9. Treat Volume-variant flags as a signal to investigate further, not as a final verdict.

Should I override the Monotony >2.0 threshold for elite athletes?

The thresholds come from professional team-sport literature, validated in football, rugby, AFL, basketball, and handball. Foster's original 2.0 threshold has held up across two decades of follow-up research. Don't tune it down — if you're seeing repeated HIGH flags on tolerant athletes, it's more useful to ask whether the athletes have been reporting RPE accurately, not whether the threshold is wrong.

How does this relate to ACWR (Gabbett 2017)?

ACWR (acute:chronic ratio) measures fast vs slow load — is the past week sharply higher than the past month? Monotony measures day-to-day variability within the past week. The two are independent and complementary. A player can have a SAFE ACWR (no spike) but a DANGER Strain (every day at the same load), or vice versa. Use both.

What if a player has a high single-session RPE but rests the rest of the week?

That player will likely show LOW monotony (high SD because of one high day) and moderate strain — exactly the right read. One big day followed by rest is a healthy pattern. The card does not penalise this.

9. Limitations — what Monotony & Strain CAN'T do

- Only sees the past 7 days. Cannot detect chronic accumulation across weeks (use ACWR / Quadrant for that).
- Depends on RPE compliance for the canonical variant. If players don't log honestly, the data is wrong.
- Does not distinguish between session types. A 600 AU recovery session and a 600 AU sprint session count the same — Foster's framework is intentionally type-agnostic.
- Monotony "—" cases (low-data, returning from injury) are not flagged — they look like missing data, not low risk. Don't infer SAFE from absence of a flag.
- Volume variant uses total distance only, ignoring intensity bands. A player who logged 6000 m at jogging pace is treated identically to one who covered 6000 m at high speed.
- Does not account for non-training stress (school, work, life, travel). For full picture, combine with subjective wellness (Saw 2016).

10. References

Foster C. (1998). Monitoring training in athletes with reference to overtraining syndrome. *Medicine and Science in Sports and Exercise*, 30(7), 1164-1168.

Foster C, Florhaug JA, Franklin J, Gottschall L, Hrovatin LA, Parker S, Doleshal P, Dodge C. (2001). A new approach to monitoring exercise training. *Journal of Strength and Conditioning Research*, 15(1), 109-115.

Saw AE, Main LC, Gastin PB. (2016). Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures. *British Journal of Sports Medicine*, 50(5), 281-291.

MicroPulse — Training Monotony & Strain Coach Guide · v1.0 · May 2026